

Competency

- To solve the daily life problems on profit and loss by using fundamental rules of profit/loss and formulae.

Learning Outcomes

- To collect the real life problems on profit and loss and solve them.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define cost price, selling price - To tell the relation of C.P., S.P. and profit % or loss% - To define marked price - To tell the formula of finding the discount - To define VAT
2.	Understanding (U)	- To find the profit/ loss amount - To calculate the profit and profit - To find the discount/ VAT amount - To calculate the rates of discount and VAT
3.	Application (A)	- To solve the verbal problems on profit and loss - To solve the verbal problems on discount and VAT
4.	High Ability (HA)	- To mathematize the daily life problems related to profit and loss and solve them. - To link various real life/ contemporary problems with discount and VAT

Required Teaching Materials/ Resources

Colourful chart-paper with required definitions and formulae, bills, VAT bills, audio/video clips related to profit and loss, projector etc.

Pre-knowledge: cost price, selling price, profit and loss

A. Profit and Loss

Teaching Activities

1. Discuss upon cost price, selling price, profit and loss of the articles like watch, mobile, books, copies, bags etc.
2. Divide the class into 5/6 groups and ask the formulae of profit amount, loss amount, profit percentage and loss percentage.
3. List out the following formulae with examples
 - (i) Profit amount = $S.P. - C.P.$
 - (ii) Loss amount = $C.P. - S.P.$
 - (iii) Profit amount = profit % of C.P.
 - (iv) Loss amount = loss % of C.P.
 - (v) $S.P. = C.P. + P\% \text{ of } C.P.$

$$(vi) \text{ S.P.} = \text{C.P.} - L\% \text{ of C.P.}$$

$$(vii) \text{ Profit percent} = \frac{\text{Profit}}{\text{C.P.}} \times 100\% \rightarrow \frac{\text{S.P.} - \text{C.P.}}{\text{C.P.}} \times 100\%$$

$$(viii) \text{ loss percent} = \frac{\text{Loss}}{\text{C.P.}} \times 100\% \rightarrow \frac{\text{C.P.} - \text{S.P.}}{\text{C.P.}} \times 100\%$$

Solution of selected problems from Vedanta Excel in Mathematics

1. **A grocer bought 20 kg of sugar at Rs 70 per kg. He sold 15 kg of sugar at Rs 75 per kg and the remaining quantity at Rs 69 per kg. Find his profit or loss percent in the transaction.**

Solution:

Here, C.P. of 20 kg of sugar = $20 \times \text{Rs } 70 = \text{Rs } 1400$

S.P. of 15 kg of sugar = $15 \times \text{Rs } 75 = \text{Rs } 1125$

Remaining quantity of sugar = $20 \text{ kg} - 15 \text{ kg} = 5 \text{ kg}$

S.P. of 5 kg of sugar = $5 \times \text{Rs } 69 = \text{Rs } 345$

S.P. of 20 kg of sugar = $\text{Rs } 1125 + \text{Rs } 345 = \text{Rs } 1470$

Since, $\text{S.P.} > \text{C.P.}$, he made a profit

Profit amount = $\text{S.P.} - \text{C.P.} = \text{Rs } 1470 - \text{Rs } 1400 = \text{Rs } 70$

Now, profit percent = $\frac{\text{Profit amount}}{\text{Cost price (C.P.)}} \times 100\% = \frac{\text{Rs } 70}{\text{Rs } 1400} \times 100\% = 5\%$

Hence, his profit percent is 5%.

2. **Mrs. Rokaya bought 2 quintals of apples in Jumla for Rs 17000. She paid Rs 15 per kg for the transportation from Jumla to Nepalgunj. 10 kg of apples is damage and she sold the remaining quantity of apples at Rs 120 per kg. Calculate her profit or loss percent in the total transaction.**

Solution:

Here, C.P. of 2 quintals i.e., 200 kg of apples = $\text{Rs } 17000$

C.P. of the apples with the transportation cost = $\text{Rs } 17000 + 200 \times \text{Rs } 15 = \text{Rs } 20000$

Saleable quantity of apples = $200 \text{ kg} - 10 \text{ kg} = 190 \text{ kg}$

S.P. of 190 kg of apples = $190 \times \text{Rs } 120 = \text{Rs } 22800$

Since, $\text{S.P.} > \text{C.P.}$, she made a profit

Profit amount = $\text{S.P.} - \text{C.P.} = \text{Rs } 22800 - \text{Rs } 20000 = \text{Rs } 2800$

Now, profit percent = $\frac{\text{Profit amount}}{\text{Cost price (C.P.)}} \times 100\% = \frac{\text{Rs } 2800}{\text{Rs } 20000} \times 100\% = 14\%$

Hence, her profit percent is 14% in the total transaction.

3. **A stationer sells 8 pencils for the cost of 10 pencils, find his gain percent.**

Solution:

Let the C.P. of 1 pencil be $\text{Rs } x$.

Then, C.P. of 10 pencils = $\text{Rs } 10x$ and C.P. of 8 pencils = $\text{Rs } 8x$

According to the question, S.P. of 8 pencils = C.P. of 10 pencils = $\text{Rs } 10x$

But C.P. of 8 pencils = $\text{Rs } 8x$. So, gain amount = $\text{Rs } 10x - \text{Rs } 8x = \text{Rs } 2x$

Now, gain percent = $\frac{\text{Gain amount}}{\text{Cost price (C.P.)}} \times 100\% = \frac{\text{Rs } 2x}{\text{Rs } 10x} \times 100\% = 20\%$

Hence, his gain percent is 20%.

4. **A stationer bought 2000 exercise books. He distributed 200 exercise books to the students from poor economical background as donation. He sold each of the remaining exercise books at Rs 10 more than the cost price of each and gained 8%, find the cost price of each exercise book.**

Solution:

Let the C.P. of each exercise book be Rs x .

Then S.P. of each exercise book is Rs $(x + 10)$

Now, C.P. of 2000 exercise books = Rs $2000x$

Saleable quantity of exercise books = $2000 - 200 = 1800$

S.P. of 1800 exercise books = Rs $1800(x + 10) = \text{Rs } (1800x + 18000)$

Gain percent = 8%

Now, S.P. = C.P. + G% of C.P. or, $1800x + 18000 = 2000x + 8\% \text{ of } 2000x$

or, $1800x + 18000 = 2160x$ or, $18000 = 360x \therefore x = \text{Rs } 50$

Hence, the cost price of each exercise book is Rs 50.

- 5. Bikash purchased 10 pens. He sold 5 pens at 25% profit and the remaining 5 pen at $16\frac{2}{3}\%$ loss. If he received Rs 625 in total, find the cost price of each pen.**

Solution:

Let the C.P. of each pen be Rs x . Then C.P. of 5 pens = Rs $5x$ and C.P. of 10 pens = Rs $10x$

S.P. of 5 pens = C.P. + P% of C.P. = $5x + 25\% \text{ of } 5x = 5x + \frac{25}{100} \times 5x = \text{Rs } \frac{25x}{4}$

S.P. of remaining 5 pens = C.P. - L% of C.P. = $5x - 16\frac{2}{3}\% \text{ of } 5x = 5x - \frac{50}{3 \times 100} \times 5x = \text{Rs } \frac{25x}{6}$

According to question, total S.P. = Rs 625

or, $\frac{25x}{4} + \frac{25x}{6} = \text{Rs } 625$ or, $\frac{125x}{12} = \text{Rs } 625 \therefore x = 60$

Hence, the cost price of each pen is Rs 60.

- 6. A dealer bought a pen-drive for Rs 500. He sold it at 10% loss. If he wanted to make a profit of 12.5% without increasing the selling price, by how much should the cost price of the pen-drive be reduced?**

Solution:

Here, C.P. of a pen-drive = Rs 500 and loss percent = 10%

Now, S.P. of pen-drive = C.P. - L% of C.P. = Rs $500 - 10\% \text{ of Rs } 500 = \text{Rs } 450$

Again, S.P. of pen-drive = Rs 450 and profit percent = 12.5%

We have, S.P. = C.P. + P% of C.P. or, Rs $450 = \text{C.P.} + 12.5\% \text{ of C.P.}$

or, Rs $450 = 1.125 \text{ C.P.} \therefore \text{C.P.} = \text{Rs } 400$

Difference between cost price = Rs $500 - \text{Rs } 400 = \text{Rs } 100$

Hence, the cost price of the pen-drive should be reduced by Rs 100 to make 12.5% profit.

- 7. A trader sold an article at 10% profit. If he sold it at 10% loss, it would yield Rs 140 less than the previous selling price; find the cost price of the article.**

Solution:

Let the cost price of an article be Rs x .

Now, S.P. of article = C.P. + P% of C.P. = Rs $x + 10\% \text{ of Rs } x = \text{Rs } 1.1x$

According to the question, new S.P. of the article = Rs $(1.1x - 140)$

We have, S.P. = C.P. - L% of C.P.

or, $1.1x - 140 = x - 10\% \text{ of } x$

or, $1.1x - 140 = 0.9x$

or, $0.2x = 140$

$\therefore x = 700$

Hence, the cost price of the article was Rs 700.

- 8. Rajesh Das bought two calculators for Rs 1000. He sold one of them at 20% profit and the other at 20% loss. If the selling prices of both the calculators are same, find the cost price of each calculator. Also, calculate his gain or loss in the total transaction.**

Solution:

Let the cost price of the one calculator be Rs x then that of the another one is Rs $(2000-x)$

From the first calculator: $C.P._1 = \text{Rs } x$, profit percent = 20%

$$\therefore S.P._1 = C.P._1 + P\% \text{ of } C.P._1 = \text{Rs } x + 20\% \text{ of } \text{Rs } x = \text{Rs } 1.2x$$

From the second calculator: $C.P._2 = \text{Rs } (1000 - x)$, loss percent = 20%

$$\begin{aligned}\therefore S.P._2 &= C.P._2 - L\% \text{ of } C.P._1 \\ &= \text{Rs } (1000 - x) - 20\% \text{ of } \text{Rs } (1000 - x) = \text{Rs } (1000 - x - 200 + 0.2x) = \text{Rs } (800 - 0.8x)\end{aligned}$$

According to the question, $S.P._1 = S.P._2$

$$\text{or, } 1.2x = 800 - 0.8x$$

$$\text{or, } 2x = 800$$

$$\therefore x = \text{Rs } 400$$

Now, the cost price of the first calculator was Rs 400 and that of the other was $\text{Rs } (1000 - 400) = \text{Rs } 600$

$$\text{Again, } S.P._1 = 1.2 \times \text{Rs } 400 = \text{Rs } 480$$

$$S.P. \text{ of two calculators} = 2 \times \text{Rs } 480 = \text{Rs } 960 \text{ and } C.P. = \text{Rs } 1000$$

Since, $C.P. > S.P.$, there is a loss

$$\text{Loss amount} = C.P. - S.P. = \text{Rs } 1000 - \text{Rs } 960 = \text{Rs } 40$$

$$\text{Now, loss percent} = \frac{\text{Loss amount}}{\text{Cost price (C.P.)}} \times 100\% = \frac{\text{Rs } 40}{\text{Rs } 1000} \times 100\% = 4\%$$

Hence, his loss percent was 4% in the total transaction.

- 9. Ajaya bought a fan and a heater for Rs 4000. He sold the fan at a profit of 25% and the heater at a loss of 5%. If he gained 7% on his total outlay, at what price did he buy each item?**

Solution:

Let the cost price of the fan be Rs x then that of the heater is Rs $(4000-x)$

From the fan: $C.P._1 = \text{Rs } x$, profit percent = 25%

$$\therefore S.P._1 = C.P._1 + P\% \text{ of } C.P._1 = \text{Rs } x + 25\% \text{ of } \text{Rs } x = \text{Rs } 1.25x$$

From the heater: $C.P._2 = \text{Rs } (4000 - x)$, loss percent = 5%

$$\begin{aligned}\therefore S.P._2 &= C.P._2 - L\% \text{ of } C.P._1 \\ &= \text{Rs } (4000 - x) - 5\% \text{ of } \text{Rs } (4000 - x) = \text{Rs } (4000 - x - 200 + 0.05x) = \text{Rs } (3800 - 0.95x)\end{aligned}$$

$$\text{Total C.P.} = \text{Rs } 4000$$

$$\text{Total S.P.} = \text{Rs } (1.25x + 3800 - 0.95x) = \text{Rs } (3800 + 0.3x) \text{ and profit percent} = 7\%$$

$$S.P. = C.P. + P\% \text{ of } C.P.$$

$$\text{or, } 3800 + 0.3x = \text{Rs } 4000 + 7\% \text{ of } 4000$$

$$\text{or, } 3800 + 0.3x = \text{Rs } 4280$$

$$\text{or, } 0.3x = \text{Rs } 480$$

$$\therefore x = \text{Rs } 1600 \text{ and } \text{Rs } (4000 - x) = \text{Rs } (4000 - 1600) = \text{Rs } 2400$$

Hence, the cost price of the fan was Rs 1600 and that of heater was Rs 2400.

10. A shopkeeper decided to make equal rate of profit in each fancy item. If he sold a jacket costing Rs 4400 for Rs 5060, at what price did he sell the shoes which was purchased for Rs 3420?

Solution:

Here,

The cost price (C.P.) of a jacket = Rs 4400 and selling price (S.P.) = Rs 5060

Now, profit percent = $\frac{\text{S. P.} - \text{C.P.}}{\text{C. P.}} \times 100\% = \frac{\text{Rs } 5060 - \text{Rs } 4400}{\text{Rs } 4400} \times 100\% = 15\%$

Again, cost price of the shoes (C.P.) = Rs 3420 and profit percent = 15%

We have, S.P. = S.P. = C.P. + P% of C.P. = Rs 3420 + 15% of Rs 3420 = Rs 3933

Hence, he sold the shoes for Rs 3933.

11. A dishonest shopkeeper has two false balances. One balance weighs 10% more while buying the goods and other weighs 10% less while selling the goods. Find the gain percent just by weighing.

Solution:

Here,

While buying the goods, he weighs 10% more.

So, C.P. of goods of worth Rs 110 = Rs 100

While selling the goods, he weighs 10% less.

So, S.P. of goods of worth Rs 100 = Rs 110

S.P. of goods costing Re 1 = $\frac{\text{Rs } 110}{100}$

Since, he has the goods of worth Rs 110.

$\therefore$ S.P. of goods of worth Re 110 = $\frac{\text{Rs } 110}{100} \times 110 = \text{Rs } 121$

Now, profit percent = $\frac{\text{S. P.} - \text{C.P.}}{\text{C. P.}} \times 100\% = \frac{\text{Rs } 121 - \text{Rs } 100}{\text{Rs } 100} \times 100\% = 21\%$

12. A crooked shopkeeper sells goods at the cost price. But his 1 kg weight weighs 900 g only. Find his gain percent.

Solution:

Let the C.P. of 1 g of goods be Rs x. Then, C.P. of 1 kg of goods = Rs 1000x and S.P. of 900 g of goods = Rs 900x

According to the question,

S.P. of 900 g = C.P. of 1 kg of goods = Rs 1000x

But C.P. of 900 g of goods = Rs 900x.

So, gain amount = Rs 1000x – Rs 900x = Rs 100x

Now, gain percent = $\frac{\text{Gain amount}}{\text{Cost price (C.P.)}} \times 100\% = \frac{\text{Rs } 100x}{\text{Rs } 900x} \times 100\% = 11\frac{1}{9}\%$

Hence, his gain percent is $11\frac{1}{9}\%$.

13 A grocer has some rice of worth Rs 3000. He sold $\frac{1}{3}$ of it with 10% loss. By how many percent must the selling price be increased for making 10% profit on the outlay?

Solution:

Here, C.P. of certain quantity of rice = Rs 3000

Total S.P. for making 10% profit = C.P. + P% of C.P. = Rs 3000 + 10% of Rs 3000 = Rs 3300

C.P. of $\frac{1}{3}$ of the rice = $\frac{1}{3}$ of Rs 3000 = Rs 1000 and loss percent = 10%

∴ S.P. of $\frac{1}{3}$ of the rice = C.P. – L% of C.P. = Rs 1000 – 10% of Rs 1000 = Rs 900

For making no any gain, S.P. of the remaining $\frac{2}{3}$ of the rice = Rs 3000 – Rs 900 = Rs 2100

For making 10% gain, S.P. of the remaining $\frac{2}{3}$ of the rice = Rs 3300 – Rs 900 = Rs 2400

Difference in S.P. = Rs 2400 – Rs 2100 = Rs 300

Now, increased percent in S.P. of remaining quantity = $\frac{\text{Rs } 300}{\text{Rs } 2100} \times 100\% = 14\frac{2}{7}\%$

Hence, the selling price of the remaining quantity of rice should be increased by $14\frac{2}{7}\%$ for making 10% profit on the outlay.

Extra Questions

1. Mr. Lama bought a second hand bike for Rs 1,10,000 and immediately he spent Rs 5,000 to repair it. Then he sold it for Rs 1,26,500. Find his profit or loss percent.

[Ans: Profit percent = 10%]

2. A grocer sold 5 kg of wheat flour at Rs 55 per kg and gained 10%. If he sold all the flour for Rs 260, what would be his gain or loss percent? [Ans: Profit 4%]

3. A man bought a hen and a duck for Rs 3,400. He sold the hen at 25% profit and then duck at 10% loss. If he gained 10% on his total outlay. At what price did he buy the hen and the duck each?

[Ans: Rs 2,000, Rs 1500]

4. Rajendra is a stationer. Once, he bought 2,000 books. Out of them he donated 200 books to a school library. He sold the remaining books with 8% profit at the rate of Rs. 120 per book. What will be the cost price of a book? Find it.

[Ans: Rs 100]

5. A shopkeeper sold a sewing machine for Rs 3,600 and made a loss of 10%. For what price should he sell it to gain 10%?

[Ans: Rs 4,400]

B. Marked price (M.P.) and Discount

Teaching Activities

- Recall cost price (C.P.) and selling price (S.P.) of an article.
- With examples, discuss on marked price (M.P.) and discount.
- Explain discount as the amount of reduction in the marked price of an article.
- Paste/show the different types of taxes in the colourful chart paper and explain with appropriate examples.
- Make clear VAT as tax levied on purchase of goods or service
- List the following formulae after discussion
 - Discount amount = M.P. – S. P.
 - Discount amount = Discount % of M.P.
 - Rate of discount = $\frac{\text{Discount amount}}{\text{M.P.}} \times 100\%$
 - S.P. = M.P. – Discount amount
 - S.P. = M.P. – Discount% of M.P. = M.P. (1 – Discount %)
 - S.P. = M.P. (1 – D₁%)(1 – D₂%) when two successive discount rates D₁% and D₂% are given.
 - VAT amount = S.P. including VAT – S. P. excluding VAT
 - VAT amount = VAT% of .P.

$$(ix) \text{ Rate of VAT} = \frac{\text{VAT amount}}{\text{S.P.}} \times 100\%$$

$$(x) \text{ S.P. with VAT} = \text{S.P.} + \text{VAT amount}$$

$$(xi) \text{ S.P. with VAT} = \text{S.P.} + \text{VAT\% of S.P.} = \text{S.P.} (1 + \text{VAT \%})$$

Solution of selected problems from Vedanta Excel in Mathematics

- 1. A retailer allows 15% discount on the marked price of an electric fan. If a customer pays Rs 2,244 with 10% VAT, find the marked price of the fan.**

Solution:

Let the marked price of the electric fan be Rs x

Then, S.P. after 15% discount = M.P. - D% of M.P. = $x - 15\% \text{ of } x = \text{Rs } 0.85x$

Again, S.P. with VAT = S.P. + VAT% of S.P.

$$\text{or, } 0.85x + 10\% \text{ of } 0.85x = \text{Rs } 2,244$$

$$\text{or, } 0.935x = \text{Rs } 2,244$$

$$\therefore x = \text{Rs } 2,400$$

Hence, the marked price of the electric fan is Rs 2,400.

- 2. A tourist paid Rs 5,610 for a carved window made up of wood with discount of 15% including 10% value added tax (VAT). How much does he get back while leaving Nepal?**

Solution:

Let the marked price of the carved window be Rs x

Then, S.P. after 15% discount = M.P. - D% of M.P. = $x - 15\% \text{ of } x = \text{Rs } 0.85x$

Also, S.P. with VAT = S.P. + VAT% of S.P.

$$\text{or, } \text{Rs } 5,610 = 0.85x + 10\% \text{ of } 0.85x$$

$$\text{or, } \text{Rs } 5,610 = 0.935x$$

$$\therefore x = \text{Rs } 6,000$$

Again, S.P. = Rs 0.85x = Rs 0.85 × 6000 = Rs 5,100

And VAT amount = 10% of Rs 5,100 = Rs 510

Hence, the tourist gets Rs 510 back while leaving Nepal.

- 3. The marked price of a cycle is Rs 5,500. After allowing certain percent of discount with 10% VAT levied, the cycle is sold at Rs 5,445, find the discount percent.**

Solution:

The marked price of a cycle = Rs 5,500, VAT percent = 10% and S.P. with VAT = Rs 5,445, discount percent = ?

Let S.P. after discount of the cycle be Rs x.

Then, S.P. with VAT = S.P. + VAT% of S.P.

$$\text{or, } \text{Rs } 5,445 = x + 10\% \text{ of } 0.85x$$

$$\text{or, } \text{Rs } 5,445 = 1.1x$$

$$x = \text{Rs } 4,950$$

Also, discount amount = M.P. - S.P. = Rs 5,500 - Rs 4,950 = Rs 550

$$\text{Again, discount percent} = \frac{\text{Discount amount}}{\text{M.P.}} \times 100\% = \frac{\text{Rs } 550}{\text{Rs } 5500} \times 100\% = 10\%$$

Hence, the required discount percent is 10%.

- 4. The mobile price is tagged Rs 5,000. If a customer gets 12% discount and adding certain percent VAT reaches as Rs 4,972, find out the VAT percent.**

Solution:

The marked price of the mobile = Rs 5,000, discount percent = 12% and S.P. with VAT = Rs 4,972, VAT percent = ?

Now,

Then, S.P. after discount = M.P. – D% of M.P. = Rs 5,000 – 12% of Rs 5,000 = Rs 4,400

Also, VAT amount = S.P. with VAT – S.P. = Rs 4,972 – Rs 4,400 = Rs 572

Again, VAT percent = $\frac{\text{VAT amount}}{\text{S.P.}} \times 100\% = \frac{\text{Rs } 572}{\text{Rs } 4400} \times 100\% = 13\%$

Hence, the required VAT percent is 13%.

- 5. After allowing 15% discount on the marked price of a camera, 15% VAT was levied and sold it. If the difference between the selling price with VAT and selling price after discount is Rs 1,122, find the marked price of the camera.**

Solution:

Let the marked price of the camera be Rs x

Then, S.P. after 15% discount = M.P. – D% of M.P. = x – 15% of x = Rs 0.85x

Again, S.P. with 15% VAT = S.P. + VAT% of S.P. = Rs 0.85x + 15% of Rs 0.85x = 0.9775x

According to the question,

S.P. with VAT – S.P. after discount = Rs 1,122

or, $0.9775x - 0.85x = \text{Rs } 1,122$

or, $0.1275x = \text{Rs } 1,122$

$\therefore x = \text{Rs } 8,800$

Hence, the marked price of the camera is Rs 8,800.

- 6. The marked price of an article is 25% above the cost price. When it is sold at a discount of 15%, there is a gain of Rs 200. Find.**
(i) The cost price of the article. (ii) The marked price of the article.

Solution:

Let the cost price (C.P.) of the article be Rs x

Then, M.P. of the article = C.P. + 25% of C.P. = Rs x + 25% of x = Rs 1.25x

Now, S.P. after 15% discount = M.P. – D% of M.P. = 1.25x – 15% of 1.25x = Rs 1.0625x

Again, profit amount = S.P. – C.P.

or, $\text{Rs } 200 = 1.0625x - x$

or, $\text{Rs } 200 = 0.0625x$

$\therefore x = \text{Rs } 3,200$

And, M.P. = Rs 1.25x = Rs 1.25 × 3200 = Rs 4,000

Hence, the cost price of the article is Rs 3,200 and its marked price is Rs 4,000.

- 7. When a pen is sold at a discount of 15%, there is a gain of Rs 10. But if it is sold at 25% discount, there is a loss of Rs 2. Find the marked price of the pen.**

Solution:

Let the marked price (M.P.) of the pen be Rs x.

According to the given first condition, discount percent = 15%, profit = Rs 10

We have, S.P. after 15% discount = M.P. – D% of M.P. = x – 15% of x = Rs 0.85x

Again, profit amount = S.P. – C.P.

or, $\text{Rs } 10 = 0.85x - \text{C.P.}$

$\therefore \text{C.P.} = \text{Rs } (0.85x - 10) \dots (i)$

According to the given second condition, discount percent = 25%, loss = Rs 2

We have, S.P. after 25% discount = M.P. – D% of M.P. = $x - 25\%$ of $x = \text{Rs } 0.75x$

Again, loss amount = C.P. – S.P.

$$\text{or, } \text{Rs } 2 = \text{Rs } (0.85x - 10) - \text{Rs } 0.75x \quad [\text{From (i)}]$$

$$\text{or, } \text{Rs } 12 = 0.1x$$

$$\therefore x = \text{Rs } 120$$

Hence, the marked price of the pen is Rs 120.

8. A shopkeeper marked the price of an article a certain percent above the cost price and he allowed 16% discount to make 5% profit. If a customer paid Rs 9,492 with 13% VAT to buy the article, by what percent is the marked price above the cost price of the article?

Solution:

Let the marked price of the article be Rs x

Then, S.P. after 16% discount = M.P. – D% of M.P. = $x - 16\%$ of $x = \text{Rs } 0.84x$

Again, S.P. with VAT = S.P. + VAT% of S.P.

$$\text{or, } 0.84x + 13\% \text{ of } 0.84x = \text{Rs } 9,492$$

$$\text{or, } 0.9492x = \text{Rs } 9,492$$

$$\therefore x = \text{Rs } 10,000$$

Hence, the marked price of the article is Rs 10,000.

Also, S.P. = Rs $0.84x = \text{Rs } 0.84 \times 10,000 = \text{Rs } 8,400$

Let C.P. of the article be Rs y

We have, S.P. = C.P. + P% of C.P.

$$\text{or, } \text{Rs } 8,400 = y + 5\% \text{ of } y$$

$$\text{or, } \text{Rs } 8,400 = 1.05y$$

$$\therefore y = \text{Rs } 8,000$$

Again, difference between M.P. and C.P. = Rs $10,000 - \text{Rs } 8,000 = \text{Rs } 2,000$

$\therefore$ The marked price of the article is above the cost price by $\frac{\text{Rs } 2,000}{\text{Rs } 8,000} \times 100\% = 25\%$

9. The marked price of an item is Rs 1,500 and 10% discount is given to make 20% profit. By what percent is the discount to be increased to get only 12% profit?

Solution:

Here, the marked price (M.P.) of the item = Rs 1,500, discount = 10 % and profit percent = 20%

Then, S.P. after discount = M.P. – D% of M.P. = Rs $1,500 - 10\%$ of Rs $1,500 = \text{Rs } 1,350$

Let the cost price (C.P.) of the item be Rs x .

We have, S.P. = C.P. + P% of C.P.

$$\text{or, } \text{Rs } 1,350 = x + 20\% \text{ of } x$$

$$\text{or, } \text{Rs } 1,350 = 1.2x$$

$$\therefore x = \text{Rs } 1,125$$

Thus, the cost price (C.P.) of the item is Rs 1,125

Again, to get only 12% profit

$$\text{S.P.} = \text{C.P.} + \text{P\% of C.P.} = \text{Rs } 1,125 + 12\% \text{ of Rs } 1,125 = \text{Rs } 1,260$$

Discount amount = M.P. – S.P. = Rs $1,500 - \text{Rs } 1,260 = \text{Rs } 240$

$$\text{Discount percent} = \frac{\text{Discount amount}}{\text{M. P.}} \times 100\% = \frac{\text{Rs } 240}{\text{Rs } 1500} \times 100\% = 16\%$$

Hence, the discount should be increased by $16\% - 10\% = 6\%$ to make only 12% profit.

- 10. The price of a watch is marked Rs 11,250. When it is sold allowing 20% discount, 20% profit is made. By what percent is the discount to be reduced to increase the profit by 3%?**

Solution:

Here, the marked price (M.P.) of the item = Rs 11,250, discount = 20 % and profit percent = 20%

Then, S.P. after discount = M.P. - D% of M.P. = Rs 11,250 - 20% of Rs 11,250 = Rs 9,000

Let the cost price (C.P.) of the item be Rs x.

We have, S.P. = C.P. + P% of C.P.

or, Rs 9,00 = x + 20% of x

or, Rs 9,000 = 1.2x

∴ x = Rs 7,500

Thus, the cost price (C.P.) of the item is Rs 7,500

Again, to get only (20 + 3)% = 23% profit

$$S.P. = C.P. + P\% \text{ of } C.P. = Rs 7,500 + 23\% \text{ of } Rs 7,500 = Rs 9,225$$

Discount amount = M.P. - S.P. = Rs 11,250 - Rs 9,225 = Rs 2,025

$$\text{Discount percent} = \frac{\text{Discount amount}}{\text{M. P.}} \times 100\% = \frac{Rs 2,025}{Rs 11,250} \times 100\% = 18\%$$

Hence, the discount should be decreased by 20% - 18% = 2% to increase the profit by 3%.

- 11. The marked price of a digital watch is Rs 6,000. Allowing 10% discount and including same percent of value added tax, the watch is sold. By how much percent is the VAT amount less than discount amount?**

Solution:

Here, the marked price (M.P.) of the digital watch = Rs 6,000, discount = 10 % and VAT = 10%

Now, Discount amount = D% of M.P. = 10% of Rs 6,000 = Rs 600

Also, S.P. after discount = M.P. - D = Rs 6,000 - Rs 600 = Rs 5,400

Again, VAT amount = VAT% of S.P. = 10% of Rs 5,400 = Rs 540

Difference between discount and VAT amounts = Rs 600 - Rs 540 = Rs 60

VAT amount is less than discount amount by $\frac{Rs 60}{Rs 600} \times 100\% = 10\%$

Hence, the VAT amount is less than discount amount by 10%.

Extra Questions

- What is the price of a bag costing Rs 2000 after allowing 15% discount? Find it. [Ans: Rs 1700]
- Mr. Ajay sold a watch for Rs 880 after allowing 20% discount, what was the marked price of the watch? [Ans: Rs 1,100]
- A shopkeeper in Nepalgunj fixed the price of a suitcase in such a way that he could gain 10% after allowing 10% discount on it. If the customer paid Rs 9900 for the suitcase, find the marked price and the selling price of the suitcase. [Ans: Rs 11,000; Rs 9000]
- What will be the price of a calculator costing Rs 600 with 13% value added tax (VAT)? [Ans: Rs 678]
- The marked price of a scooter is Rs 2,40,000. If the shopkeeper allows 15% discount and levies 13% value added tax, how much should a customer have to buy the scooter? Find it. [Ans: Rs 2,30,520]
- A cycle was sold after allowing 20% discount on the marked price and levying 10% VAT. If the customer got Rs 555 as the discount, how much VAT amount was levied on the cost of the cycle? Find it. [Ans: Rs 222]